

Mahak Singh

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Summary

Computer Science undergraduate with experience integrating computer vision into a production AR application. Skilled in Java and Spring Boot through backend projects involving distributed microservices architecture and role-based REST APIs, with database persistence, authentication, and transactional integrity. Strong foundation in Data Structures, Algorithms, and Object-Oriented Programming.

Education

B.Tech in Computer Science
ABES Engineering College

Oct 2023 – May 2027
CGPA: 7.7/10

Technical Skills

- **Programming Languages:** Java, Python
- **Backend:** Spring Boot, REST APIs
- **Database:** SQL (MySQL, PostgreSQL), MongoDB, Hibernate/JPA
- **Computer Vision:** OpenCV, MediaPipe
- **Tools:** Git, GitHub, Postman, Jira
- **Core CS Concepts:** Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, OOPs

Experience

Software Developer Intern, Senraa / Evaa Group

Oct 2025 – Apr 2026

- Integrated computer vision tools into an AR-based virtual try-on product, covering image overlay, silhouette detection, and live camera feed processing.
- Engineered the backend logic for a weighted image scoring system in Python (silhouette framing, brightness, sharpness, posture) that scores input quality out of 100 and automatically rejects images below the 70 threshold, reducing low-quality image submissions reaching the pipeline.
- Evaluated CatVTON on 100+ images across different silhouettes for garment fitting quality, and used IDM-VTON for mask generation; separately tested the PARE model on 10 videos for 3D human mesh generation as a potential improvement to fitting accuracy.

Projects

AI-Powered Health Recommendation System (Java, Spring Boot, Keycloak, RabbitMQ) | **GitHub** | **Demo** 2026

- Architected a backend system converting raw wearable data across 10 activity types and dynamic custom metrics into personalized, AI-generated health recommendations, validated by posting 100+ test activity records through the pipeline.
- Designed a 6-service distributed microservices architecture (Config Server, Eureka Service Registry, API Gateway, User/Activity/AI services) with polyglot persistence across 2 databases — PostgreSQL for user identity, MongoDB for health metrics and AI-generated insights.
- Decoupled activity ingestion from AI processing using RabbitMQ async messaging, removing 2-3 seconds of Gemini API latency from the synchronous request path.
- Integrated Keycloak for OAuth2/JWT authentication, validated once at the API Gateway instead of in each of the 3 domain services, centralizing security across the system.

ShopVerse - E-Commerce Backend Platform (Java, Spring Boot, PostgreSQL, JWT) | **GitHub** | **Demo** 2026

- Built a production-grade e-commerce REST API of 44 endpoints across 7 controllers, implementing Spring Security and JWT (JJWT) cookie authentication to enforce 3-tier Role-Based Access Control (RBAC) across Customers, Sellers, and Admins.
- Integrated dual payment gateways (Stripe PaymentIntents & PayPal Express) with real-time stock validation and an automated coupon engine, executing order confirmation within atomic (@Transactional) database boundaries.
- Designed an audit-safe database schema to preserve historical order records during user address deletions, powering multipart file upload services and administrative sales analytics APIs.

Activities & Achievements

- Finalist – IBM Prarambh Hack Challenge 2025 (Top among 1,200+ teams).
- Certified in Spring 6 and Spring Boot 3 (Telusko).
- Certified in AWS Cloud Foundations.